

DC-OPS

On-Device Data Center Operations Assistant

Qualcomm x Meta ExecuTorch Hackathon — June 2026

Real-time AI-powered server rack inspection

Running entirely on Samsung Galaxy S25 Ultra — Snapdragon 8 Elite NPU



THE PROBLEM

1000+

Server racks built every
week

~4000+

Total weekly volume over
the next few years

80%

Of racks are installed
incorrectly the first time they
are built

Cables in the wrong place

A rack is a dense mesh of cables linking trays, assembled at volume by labor crews constantly cross-referencing a paper assembly guide — slow and error-prone.



THE SOLUTION

Point your phone at any server rack.

AI identifies every component in real-time.

All processing happens on-device — zero cloud.

16 Component Classes

Compute trays, ports, LEDs, cables, fans, drives, PSUs, DPUs, and more

RAG Knowledge Overlay

Tap any detection → specs, troubleshooting, LED meanings, maintenance steps

CPU vs NPU Toggle

Live comparison: XNNPACK (CPU)
↔ QNN HTP (Snapdragon NPU) with FPS/latency

WHY ON-DEVICE?

0ms

Cloud Latency

No round-trip — inference
directly on Hexagon NPU

100%

Offline

Works in air-gapped rooms
with zero connectivity

**0
bytes**
Data Leaked

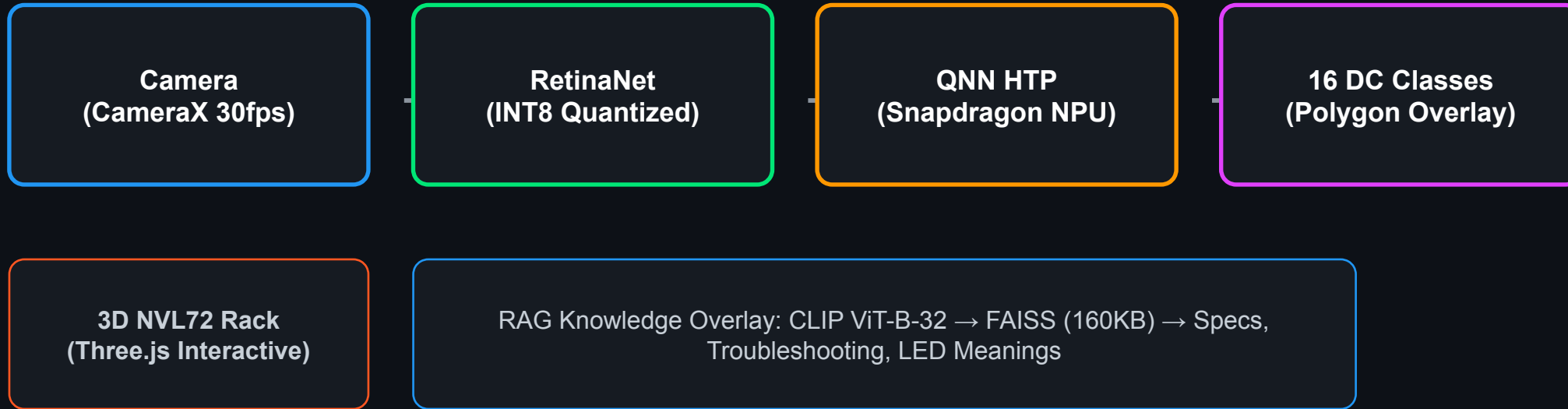
Serial numbers and infra
data never leave the device

4-5x

Power Efficient

NPU vs CPU — longer
battery life for extended
inspections

ARCHITECTURE



Android App → ExecuTorch Runtime → QAIRT SDK → Snapdragon NPU

TECH STACK

Detection (NPU)	RetinaNet-ResNet50-FPN	QNN HTP, INT8, 36MB
Detection (CPU)	YOLOv8n-seg v3	XNNPACK, FP32, 13MB
RAG Retrieval	CLIP ViT-B-32 + FAISS	160KB index, on-device
Runtime	ExecuTorch 1.4	QNN backend for SM8750
App	Kotlin + CameraX	ExecuTorch AAR (3.4MB)
Device	Galaxy S25 Ultra	Snapdragon 8 Elite, Hexagon v79
3D Visualization	Three.js NVL72 Model	Interactive rack explorer

TRAINING PIPELINE

PHASE 1 — Bootstrap

BrightData Web Unlocker scraped 300+ data-center images → auto-labeled with Grounding DINO + SAM (3,045 polygons)

PHASE 2 — 2,036 human-labeled images from 4 Roboflow datasets (CC BY 4.0) · used for final models

Server Vision

1,293 imgs

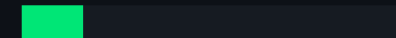
30 classes (CPU, DIMM, fans, PSUs, drives)



PC Ports

327 imgs

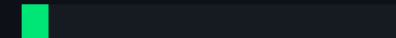
4 classes (HDMI, RJ45, USB-A, USB-C)



Ports (ym)

138 imgs

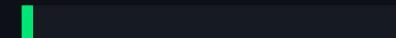
7 classes (DP, ethernet, power, USB)



Server Detection

54 imgs

9 classes (SFP, XFP, patch panels)



50+ source classes → mapped to 16 DC-Ops classes · Augmentations: Moiré, screen glare, brightness, color tint, scanlines

⚡ **Powered by BrightData — web data infrastructure for the bootstrap dataset**

MODEL PERFORMANCE

Fine-tuned YOLOv8n-seg (Ultralytics), adapted for 16 data center component classes

0.749

mAP50 (box)

Fine-tuned YOLOv8n-seg

0.85

Final Loss

RetinaNet (moiré aug)

6.4ms

Inference

Per frame on T4 GPU

compute tray

0.924

network port

0.912

power shelf

0.832

LED indicator

0.808

server rack

0.683

drive bay

0.912

DPU

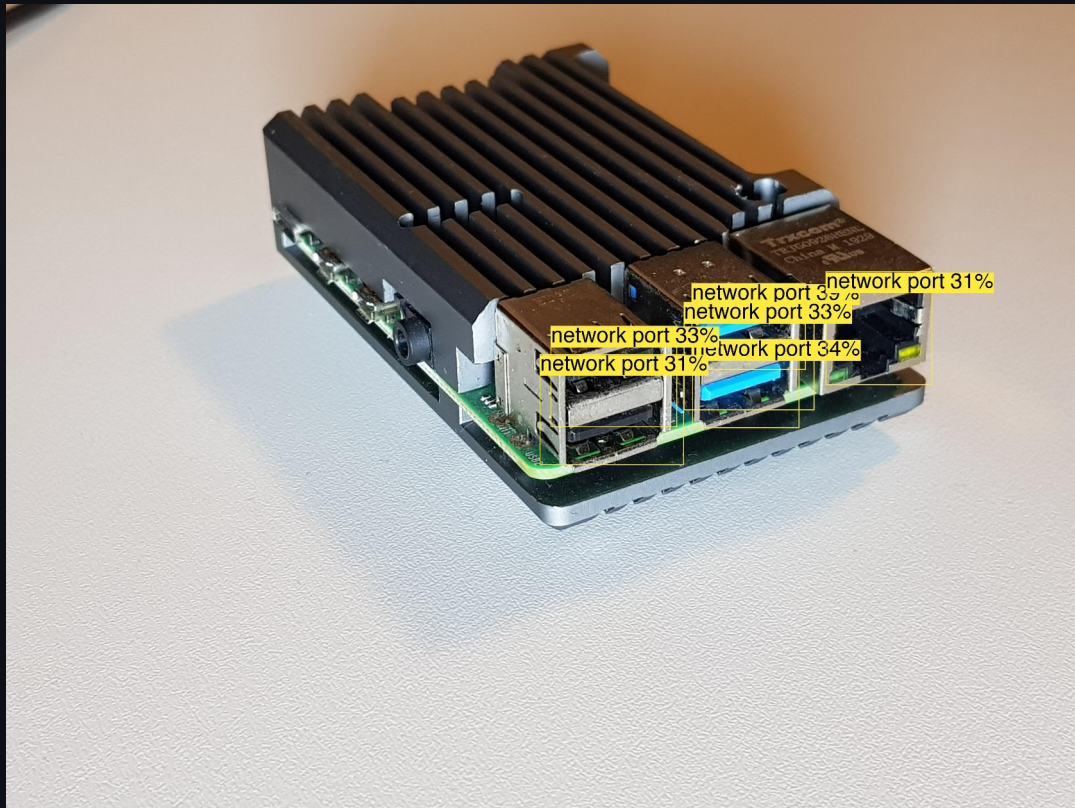
0.808

cooling manifold

0.833

RETINANET IN ACTION

Real detections on a Raspberry Pi & USB hub — mirrors our physical demo prop



All USB + RJ45 ports detected as 'network port' — fully on-device

RAG KNOWLEDGE OVERLAY

Detect → Embed → Retrieve → Display

COMPUTE TRAY

92%

⚙️ 2x Grace CPU, 4x B200 GPU, 192GB HBM3e



Green LED = healthy and operational



Amber LED = check BMC log, verify coolant



Red LED = critical hardware failure



Maintenance: check cold plate contact,
verify NVLink connector seating

- ▶ 80 knowledge chunks
- ▶ CLIP ViT-B-32 embeddings
- ▶ FAISS vector index (160KB)
- ▶ NVL72-specific documentation
- ▶ All on-device, zero cloud

CPU vs NPU COMPARISON

Backend	XNNPACK (CPU)	QNN HTP (NPU)
Model	YOLOv8n-seg	RetinaNet
Size	13 MB	36 MB
Quantization	FP32	INT8
Hardware	ARM Kryo cores	Hexagon HTP v79
Latency †	~15-30 ms	~3-5 ms
Throughput †	~30-60 FPS	~100+ FPS
Power †	Baseline	4-5x efficient

† Estimated from Qualcomm QNN HTP benchmarks — not yet measured on our specific model. Measured FPS shown live in-app.

WHAT'S NEXT

Fine-grained Port Classification

Split 'network port' into HDMI, USB-A, USB-C, RJ45, SFP individually

LED State Detection

Classify LED colors (green/amber/red) and blink patterns for health scoring

On-Device LLM Integration

LLaMA 3.2 1B via ExecuTorch for natural language troubleshooting Q&A

Audit Log + Reporting

SQLite database of all inspections, exportable via secure channel

Multi-Rack Mapping

Combine depth estimation + detection for spatial rack layout understanding

DC-OPS

On-Device Data Center Operations Assistant

GitHub

<https://github.com/abhijitbetigeri/DC-Ops>

Hugging Face

<https://huggingface.co/datasets/abhijitbetigeri/dc-ops-dataset>

Models

RetinaNet QNN HTP (.pte) · YOLOv8n-seg (.pte) · CLIP + FAISS RAG index

Built with ExecuTorch + Qualcomm QNN HTP + Snapdragon 8 Elite

Thank you!